Three Exact Four-Pole Reductions for the Standard Five-Cycle Double Cover Conjecture

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Resolution status. The standard five-cycle double cover conjecture remains open. This note proves three positive substitution reductions. It proves neither the conjecture nor its negation, and it makes no claim about the orientable five-cycle double cover conjecture.

AI-use disclosure. OpenAI Codex agents, under Atharva Vaidya's direction, selected and tested the poles, found the finite label certificates, wrote the checking programs, and drafted substantial portions of this manuscript. The certificate tables and source code are supplied for independent human checking. AI assistance is not peer review. No human specialist has yet certified the literature comparison or accepted scholarly responsibility for a submission.

Abstract

Write $D_5 = \binom{[5]}{2} \subseteq \mathbb{F}_2^5$. A standard five-cycle double cover of a cubic graph is equivalently a D_5-label on every edge whose three incident labels xor to zero at each vertex. This makes a cubic four-pole an exact finite boundary relation.

We give three human-checkable positive relations. First, the four-pole obtained by deleting adjacent vertices from the Petersen graph extends every boundary word that is a permutation of (q, q, r, r), for arbitrary $q, r \in D_5$. A 13-row certificate covers the three intersection types of q, r and all port placements. Second, deleting adjacent vertices from the cube gives a six-vertex full-boundary four-pole. It is minimum-order among connected simple terminal-distinct full-boundary poles. Third, a four-pole obtained by deleting adjacent vertices from the Heawood graph has the full boundary relation: it extends every ordered $(A_0, A_1, A_2, A_3) \in D_5^4$ with xor zero. A 10-row certificate, together with the global S_5 action, covers all 640 admissible words.

It follows that replacing two independent edges of any FiveCDC-positive cubic graph by any of the three poles preserves FiveCDC for every bijection from the four exposed endpoints to the four ports. We prove separately that the operation preserves simplicity and bridgelessness under the stated hypotheses. Thus none of these substitutions can manufacture a minimum counterexample from a smaller positive graph. The Petersen certificate establishes only the repeated-pair interface needed by this insertion, whereas the cube and Heawood certificates prove the complete xor-zero boundary relation.

The tempting extension to all bridgeless bipartite cubic graphs is false: deleting adjacent vertices from $K_{3,3}$ gives $K_{2,2}$, whose ordered boundary relation has 580 of the 640 admissible words and misses one 60-word orbit. This also proves the order-six lower bound for the cube pole. We give a direct proof of that obstruction. All claims here are local positive reductions or scope boundaries, not a resolution of FiveCDC.

Contents

1 Status, conventions, and prior-art boundary

A standard five-cycle double cover of a finite graph is a family of at most five Eulerian edge-subsets in which every edge occurs exactly twice. Equivalently, one may require exactly five indexed subsets and allow empty members. An Eulerian edge-subset need not be connected or 2-regular. Recent papers continue to state FiveCDC as open [2, 1]; special cases are known [3].

Graphs in the three pole reductions below are finite and undirected. The input and output graphs are simple and cubic. The pair-label equivalence itself also works for multigraphs if parallel edges are kept as distinct edge objects and a loop contributes two incidences at its endpoint. The finite pole certificates are for the displayed simple cores only; no computational conclusion is extrapolated to arbitrary multigraphs.

Máčajová, Mazzuoccolo, and Tabarelli introduced the relevant CDC-colouring language for multipoles and the ten boundary types of a four-pole [4]. That boundary framework, including its use in minimum-counterexample reductions, is prior work and is not claimed here. The present contribution is the three explicit extension certificates and the resulting insertion lemmas. We have not completed a specialist search capable of deciding whether any particular pole theorem, or a more general theorem implying it, already occurs in the literature. Accordingly no novelty or priority claim is made. The authors of the four-pole literature should be consulted before public submission.

The orientable five-cycle double cover conjecture adds a direction condition: every edge must be traversed once in each direction. No orientation variables or directed compatibility conditions occur below, so none of the results imply an orientable statement.

2 Pair labels and exact boundary relations

Put

$$[5] = \{0, 1, 2, 3, 4\}, \qquad D_5 = \binom{[5]}{2} \subseteq \mathbb{F}_2^5.$$

We identify a pair with its weight-two characteristic vector.

Proposition 2.1 (pair-label equivalence)**.** *For a finite graph G, the following are equivalent.*

1. *There are five Eulerian edge-subsets $C_0, \dots, C_4$ containing every edge exactly twice.*
2. *Every edge has a label $L(e) \in D_5$, and the xor of the labels on the edge incidences at every vertex is zero.*

Proof. Given the five subsets, set

$$L(e) = \{i : e \in C_i\}.$$

Exact double coverage gives $|L(e)| = 2$. In coordinate i, the vertex xor equation is exactly the even-degree condition for C_i. Conversely, the coordinate supports of a pair labelling are Eulerian by the vertex equations, and every edge belongs to its two label coordinates. □

At a cubic vertex, three labels $A, B, C \in D_5$ have $A \oplus B \oplus C = 0$ if and only if they are the three edges of a triangle in K_5.

Definition 2.2. A *terminal-distinct cubic four-pole* is a proper graph with four specified degree-two vertices, called ports, and every other vertex of degree three. Completing each port with one semiedge makes every vertex cubic. Its ordered boundary relation $\mathcal{R}_5(P)$ is the set of four semiedge-label words that extend to a D_5-labelling of all proper edges with xor zero at every completed vertex.

Lemma 2.3 (boundary parity)**.** *Every* $(A_0, A_1, A_2, A_3) \in \mathcal{R}_5(P)$ *satisfies*

$$A_0 \oplus A_1 \oplus A_2 \oplus A_3 = 0. \tag{2.1}$$

Proof. Xor all internal vertex equations. Each proper edge occurs at two endpoints and cancels. Only the four semiedge labels remain. □

Lemma 2.4 (ten full-boundary types)**.** *The 640 ordered words in* D_5^4 *satisfying* (2.1) *form ten orbits under the simultaneous* S_5 *action on coordinates.*

Proof. View the four pair labels as four ordered edges of a loopless multigraph on the five coordinate vertices. Equation (2.1) says that every coordinate has even degree. Ignoring edge order, the possibilities are four parallel edges, two doubled adjacent pairs, two doubled disjoint pairs, or one simple four-cycle. The first has one ordered-edge orbit. Each of the other three shapes has three orbits, determined by the induced partition or opposite-position pair among the four boundary positions. Hence there are $1 + 3 + 3 + 3 = 10$ orbits. Their sizes in the table of Section 5 sum to 640. □

3 A generic insertion lemma

Let P be a connected terminal-distinct cubic four-pole with ordered port set T. Let G be a cubic graph and let

$$e = ab, \qquad f = cd$$

be independent edges. For an arbitrary bijection $\pi : \{a, b, c, d\} \to T$, define

$$G \star_\pi P = \big(G - \{e, f\}\big) \cup P \cup \{z\pi(z) : z \in \{a, b, c, d\}\}. \tag{3.1}$$

There is no restriction on which ports receive the two ends of one old edge.

Definition 3.1. The pole P is *repeated-pair universal* if, for all $q, r \in D_5$, every ordered placement of the multiset (q, q, r, r) belongs to $\mathcal{R}_5(P)$. It is *full-boundary* if equality holds in Lemma 2.3.

Every full-boundary pole is repeated-pair universal.

Theorem 3.2 (positive insertion)**.** *If G has a standard FiveCDC and P is repeated-pair universal, then $G \star_\pi P$ has a standard FiveCDC for every port bijection π.*

Proof. Take a D_5-labelling of G, and let $q = L(e)$ and $r = L(f)$. Label the new joining edges at a, b by q and those at c, d by r. In the ordered port convention this is some placement of (q, q, r, r), so it extends through P. Every old vertex retains the same local equation it had in G. Proposition 2.1 converts the resulting labelling into a standard FiveCDC. □

Lemma 3.3 (simplicity and bridgelessness)**.** *Suppose G is connected, simple, cubic, and bridgeless; e, f are independent; and the proper core of P is connected, simple, and bridgeless. Then $G \star_\pi P$ is connected, simple, cubic, and bridgeless for every π.*

Proof. Cubicity and simplicity follow directly from fresh, terminal-distinct pole vertices and four distinct exposed endpoints.

Put $H = G - \{e, f\}$. Every component of H contains at least two exposed endpoints: a component containing exactly one would have a one-edge boundary in G, making that edge a bridge. Joining all components to the connected pole makes the output connected.

Every proper pole edge lies on a circuit inside the bridgeless core. For a joining edge $z\pi(z)$, choose another exposed endpoint z' in the same component of H. A z-to-z' path in that component, the joining edge at z', and a path in the pole from $\pi(z')$ to $\pi(z)$ close a circuit through $z\pi(z)$.

Finally let g be an old edge of H. If it is not a bridge in its H-component, it already lies on a circuit. Otherwise deleting g splits that component into shores X, Y. Each shore contains an exposed endpoint; if X did not, then $\delta_G(X) = \{g\}$, contradicting bridgelessness of G. Paths in X, Y, two joining edges, and a path through the pole close a circuit through g. Every output edge is therefore on a circuit. □

Corollary 3.4 (minimal-counterexample pruning)**.** *If a graph made by the insertion* (3.1) *is a FiveCDC counterexample, then the smaller graph G is already a counterexample. In particular, neither pole certified below can create a minimum counterexample from a smaller FiveCDC-positive graph.*

4 The Petersen repeated-pair certificate

Use the Petersen graph6 record

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ICOf@pSb?
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and delete the adjacent vertices $0, 3$. Keep the original vertex names. The ordered ports are

$$(6, 9, 7, 8), \tag{4.1}$$

and the ordered proper edges are

$$\begin{aligned}&(1,4),(1,6),(1,8),(2,5),(2,6),\\&(2,7),(4,7),(4,9),(5,8),(5,9).\end{aligned} \tag{4.2}$$

The proper core is simple, connected, bridgeless, and has girth five.

Theorem 4.1 (Petersen repeated-pair extension)**.** *For all $q, r \in D_5$, every ordered placement of (q, q, r, r) on the ports* (4.1) *extends through the Petersen pole.*

Proof. Under S_5, an ordered pair q, r is of one of three kinds: equal, distinct with one common coordinate, or disjoint. Representatives are $(01, 01)$, $(01, 02)$, and $(01, 23)$. For unequal labels there are six placements of the two copies of each label. The following 13 rows give the boundary labels in port order (4.1), followed by labels on the ten proper edges in order (4.2).

Port labels	Proper-edge labels
01 01 01 01	01 02 12 01 12 02 12 02 02 12
01 01 02 02	01 02 12 02 12 01 12 02 01 12
01 02 01 02	02 03 23 34 13 14 04 24 03 04
01 02 02 01	34 03 04 01 13 03 23 24 14 04
02 01 01 02	34 03 04 02 23 03 13 14 24 04
02 01 02 01	01 03 13 34 23 24 04 14 03 04
02 02 01 01	02 01 12 01 12 02 12 01 02 12
01 01 23 23	01 02 12 01 12 02 03 13 13 03
01 23 01 23	01 02 12 01 12 02 12 02 13 03
01 23 23 01	01 02 12 01 12 02 03 13 02 12
23 01 01 23	01 02 12 01 03 13 03 13 13 03
23 01 23 01	01 02 12 01 03 13 12 02 02 12
23 23 01 01	01 02 12 01 03 13 03 13 02 12

Every entry has weight two. At each of the eight completed pole vertices, the xor of its three incident labels is zero; this is checked directly from (4.1)–(4.2). Applying a simultaneous coordinate permutation preserves these equations. The three intersection types and the displayed placements exhaust all q, r, proving the theorem. □

Remark 4.2 (certified scope). The 13-row theorem proves exactly the interface required by Theorem 3.2. It is not presented here as a classification of all 640 xor-zero boundary words. The Heawood theorem below has that stronger, full-boundary conclusion.

5 The Heawood full-boundary certificate

The Heawood four-pole has graph6 record

`KhEGHC@AI?_P`.

Its ordered ports are

$$(0, 3, 8, 11), \tag{5.1}$$

and its ordered proper edges are

$$\begin{aligned}&(0,1),(1,2),(2,3),(3,4),(0,5),(4,5),(5,6),(2,7),\\&(6,7),(7,8),(4,9),(8,9),(1,10),(9,10),(6,11),(10,11).\end{aligned} \tag{5.2}$$

This proper core is simple, connected, bridgeless, and has girth six.

Theorem 5.1 (Heawood full boundary)**.** *For every $A_0, A_1, A_2, A_3 \in D_5$ satisfying*

$$A_0 \oplus A_1 \oplus A_2 \oplus A_3 = 0,$$

the Heawood four-pole has a D_5-labelling with those labels on the ordered ports (5.1)*.*

Proof. The table contains one representative of each orbit in Lemma 2.4. The middle column gives labels on the ports (5.1); the last column gives labels on the sixteen edges (5.2).

Type	Port labels	Proper-edge labels
AA	02 02 02 02	01 02 01 12 12 02 01 12 02 01 01 12 12 02 12 01
AT2	02 02 03 03	01 02 03 23 12 02 01 23 03 02 03 23 12 02 13 01
AT3	02 03 02 03	01 02 04 34 12 14 24 24 34 23 13 03 12 01 23 02
AT4	02 03 03 02	01 02 01 13 12 01 02 12 01 02 03 23 12 02 12 01
T2T2	02 02 13 13	01 02 01 12 12 02 01 12 02 01 01 03 12 13 12 23
T2T3	02 03 12 13	01 02 01 13 12 01 02 12 01 02 03 01 12 13 12 23
T2T4	02 03 13 12	01 02 04 34 12 13 23 24 12 14 14 34 12 13 13 23
T3T3	02 13 02 13	01 02 01 03 12 02 01 12 02 01 23 12 12 13 12 23
T3T4	02 13 03 12	01 02 03 01 12 02 01 23 03 02 12 23 12 13 13 23
T4T4	02 13 13 02	01 02 01 03 12 01 02 12 24 14 13 34 12 14 04 24

Every displayed label has weight two. Completing the ports in order, the xor of the three incident labels is zero at each of the twelve vertices. Thus every row is a direct positive certificate.

The orbit sizes in the displayed order are

$$10, 60, 60, 60, 30, 120, 120, 30, 120, 30,$$

whose sum is 640. Lemma 2.4 proves that these are all admissible boundary orbits. A simultaneous coordinate permutation of a row preserves label weight and every vertex xor equation, so all 640 words extend. □

Corollary 5.2 (Petersen and Heawood reductions)**.** *The insertion theorem and the bridgelessness lemma apply to the Petersen and Heawood poles, with every one of the* 4! *port bijections. Neither construction is a graph cover or lift.*

6 A sharp scope boundary: $K_{3,3}$

The Heawood graph is cubic, bipartite, and bridgeless. It is therefore natural to ask whether deleting the ends of any edge from any bridgeless bipartite cubic graph always gives a full-boundary four-pole. The smallest example shows that this statement is false.

Proposition 6.1 ($K_{3,3}$ obstruction)**.** *Delete adjacent vertices from $K_{3,3}$. The proper four-pole is $K_{2,2}$. Order its ports with the two vertices of one bipartition first and the two vertices of the other bipartition second. Its boundary relation contains 580 of the 640 xor-zero words and misses exactly the 60-word orbit represented by*

$$(02, 02, 03, 03). \tag{6.1}$$

In particular it is not full-boundary.

Proof. The exact count and uniqueness of the missing orbit follow from the standard-library finite replay described in Section 8. Here is a direct proof that (6.1) does not extend.

Let the two first-part vertices be x_0, x_1, and let y be either second-part vertex. At x_0, the two proper-edge labels have xor 02. Therefore they are

$$\{0, t_0\}, \ \{2, t_0\}$$

in some order, where $t_0 \notin \{0, 2\}$. The same reasoning at x_1 shows that its two proper-edge labels are

$$\{0, t_1\}, \ \{2, t_1\}$$

in some order, with $t_1 \notin \{0, 2\}$.

The two proper edges incident with y, one from each first-part vertex, must have xor 03. If their entries from $\{0, 2\}$ agree, their xor is contained in $\{t_0, t_1\}$, which cannot contain coordinate 0. If those entries differ, their xor contains both 0 and 2, so it cannot equal 03. This contradicts the vertex equation at y. □

Remark 6.2. Direct exact checks give full boundary for the adjacent-deletion poles of the cube, Heawood, Möbius–Kantor, Pappus, and Desargues graphs, while $K_{3,3}$ gives the obstruction above. These named examples do not support a theorem for all larger bipartite cubic graphs. A proof would need an additional structural hypothesis, and no such hypothesis is claimed here.

7 The minimum full-boundary pole: the cube

Use the cube edge set

$$(0,1),(0,3),(0,4),(1,2),(1,7),(2,3),$$
$$(2,6),(3,5),(4,5),(4,7),(5,6),(6,7),$$

and delete the adjacent vertices $0,1$. The ordered ports are

$$(2,3,4,7), \tag{7.1}$$

and the ordered proper edges are

$$(2,3),(2,6),(3,5),(4,5),(4,7),(5,6),(6,7). \tag{7.2}$$

The six-vertex proper core is simple, connected, and bridgeless.

Theorem 7.1 (minimum full-boundary pole)**.** *The cube pole* (7.1)–(7.2) *has the full boundary relation. Moreover, no connected simple terminal-distinct cubic full-boundary four-pole has fewer than six proper vertices.*

Proof. The table has one representative of each orbit in Lemma 2.4. Its last column gives labels on the seven proper edges in the order (7.2).

Type	Port labels	Proper-edge labels
`AA`	`02 02 02 02`	`12 01 01 12 01 02 12`
`AT2`	`02 02 03 03`	`01 12 12 02 23 01 02`
`AT3`	`02 03 02 03`	`23 03 02 12 01 01 13`
`AT4`	`02 03 03 02`	`01 12 13 34 04 14 24`
`T2T2`	`02 02 13 13`	`24 04 04 14 34 01 14`
`T2T3`	`02 03 12 13`	`23 03 02 24 14 04 34`
`T2T4`	`02 03 13 12`	`23 03 02 12 23 01 13`
`T3T3`	`02 13 02 13`	`03 23 01 12 01 02 03`
`T3T4`	`02 13 03 12`	`23 03 12 02 23 01 13`
`T4T4`	`02 13 13 02`	`03 23 01 12 23 02 03`

At each of the six completed vertices, the xor of its three incident labels is zero. The orbit sizes are the same

$$10,60,60,60,30,120,120,30,120,30$$

as in the Heawood table, and sum to all 640 admissible boundary words. Coordinate permutations preserve every equation, proving full boundary.

For minimality, let the proper core have n vertices and m edges. Four vertices have degree two and all others degree three, so

$$2m = 3n - 4.$$

Thus n is even and $n \geq 4$. At $n = 4$, all four vertices are ports, so a connected simple core is 2-regular and hence is C_4. This is the $K_{2,2}$ pole of Proposition 6.1, which is not full-boundary. The displayed order-six pole therefore has minimum order. □

Corollary 7.2 (minimum full-boundary insertion)**.** *The insertion theorem and the bridgelessness lemma apply to the cube pole for every one of the* 4! *port bijections. This construction is not a graph cover or lift.*

Remark 7.3. The cube theorem is the smallest full-boundary certificate in the stated simple terminal-distinct class. The Heawood theorem remains distinct in providing a girth-six proper core. Neither pole is known to be unavoidable in a minimum counterexample.

8 Replay, trust boundary, and limitations

The compact source artifacts are:

```
scratch/fivecdc-petersen-four-pole-extension-theorem-20260728.md
scratch/verify_petersen_four_pole_extension_theorem_20260728.py
scratch/fivecdc-heawood-full-four-pole-theorem-20260728.md
scratch/verify_heawood_four_pole_full_boundary_20260728.py
scratch/heawood-four-pole-full-boundary-result-20260728.json
scratch/heawood-four-pole-SHA256SUMS-20260728.txt
scratch/fivecdc-minimum-full-boundary-cube-four-pole-20260728.md
scratch/verify_cube_four_pole_full_boundary_20260728.py
scratch/verify_k33_four_pole_boundary_20260728.py
```

Run, from the repository root:

```
python3 scratch/verify_petersen_four_pole_extension_theorem_20260728.py
python3 scratch/verify_heawood_four_pole_full_boundary_20260728.py
python3 scratch/verify_cube_four_pole_full_boundary_20260728.py
python3 scratch/verify_k33_four_pole_boundary_20260728.py
sha256sum -c scratch/heawood-four-pole-SHA256SUMS-20260728.txt
```

The Petersen checker verifies the 13 displayed rows and checks all 100 ordered choices of q, r, all distinct placements encountered by that ordered enumeration, all 120 coordinate permutations, and every internal parity equation. The Heawood checker is separately written and uses only the Python standard library. It parses the graph6 record, verifies the frozen edge table, simplicity, connectivity, bridgelessness, and girth, checks every displayed row, enumerates all 10^4 boundary words, retains the 640 xor-zero words, and verifies exact coverage by the ten coordinate orbits. Its frozen JSON must equal the fresh output. The $K_{3,3}$ checker exhausts all 10^4 proper-edge labellings of $K_{2,2}$, obtains exactly 580 boundary words, and verifies that the 60 missing admissible words are precisely the S_5-orbit of $(02, 02, 03, 03)$. Its SHA-256 is

`1601e5eaeac1a3f059ab1ab219a05c7fec3e53de9679ac5ef54319fe14e908cf`.

The cube checker separately verifies the literal six-vertex core, edge-by-edge bridgelessness, all ten certificate rows, the ten orbit sizes, and exact 640-word coverage.

These are positive finite certificates. They do not rely on a SAT solver, and no UNSAT proof trace is relevant. A human can check each row by placing its edge labels on the displayed edge list and xoring the three labels at every completed vertex. The $K_{3,3}$ negative statement has both the two-line structural obstruction in Proposition 6.1 and the complete finite boundary checker.

The limitations are decisive.

1. None of the three certified insertions supplies a decomposition of every bridgeless graph into reducible pieces.

2. A minimum counterexample may contain none of the displayed poles.
3. Full boundary for the cube and Heawood poles does not imply full boundary for general four-poles, bipartite poles, or high-connectivity poles.
4. The results concern standard Eulerian edge-subsets, not orientable cycle double covers.
5. No literature novelty claim is made without specialist review.
6. FiveCDC remains unresolved.

AI-use statement

The first-page disclosure is part of this manuscript and must remain in any derivative circulation. OpenAI Codex agents materially contributed to the mathematical search, finite certificates, checking software, literature triage, and prose. The human-readable proofs and complete tables are included so that a reader need not accept an opaque model output. A human author should independently verify the tables, rerun the checkers, complete the prior-art comparison, and accept ordinary scholarly responsibility before submission. AI systems must not be listed as authors.

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